

n -step transition probabilities

Chapman-Kolmogorov Equation

Let $S = \{1, 2, \dots, m\}$. Then, the n -step transition probabilities can be obtained using the following equation:

$$p_{ij}^{(n)} = \sum_{r=1}^m p_{ir}^{(k)} p_{rj}^{(n-k)}$$

Proof:

$$\begin{aligned} p_{ij}^{(n)} &= P(X_n = j \mid X_0 = i) \\ &= \sum_{r=1}^m P(X_n = j, X_k = r \mid X_0 = i) \\ &= \sum_{r=1}^m P(X_n = j \mid X_k = r, X_0 = i) P(X_k = r \mid X_0 = i) \\ &= \sum_{r=1}^m P(X_k = r \mid X_0 = i) P(X_n = j \mid X_k = r) \\ &= \sum_{r=1}^m p_{ir}^{(k)} p_{rj}^{(n-k)} \end{aligned}$$

2-step transition probabilities in matrix notation

Using Chapman-Kolmogorov Equation, we get

$$p_{ij}^{(2)} = \sum_{r=1}^m p_{ir} p_{rj}$$

Let the one-step transition probability matrix be denoted by P . Then

$$P^2 = P \cdot P$$

$$= \begin{pmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{pmatrix} \begin{pmatrix} p_{11} & \cdots & p_{1m} \\ \vdots & \ddots & \vdots \\ p_{m1} & \cdots & p_{mm} \end{pmatrix}$$

$$= \begin{pmatrix} \sum_r p_{1r}p_{r1} & \cdots & \sum_r p_{1r}p_{rm} \\ \vdots & \ddots & \vdots \\ \sum_r p_{mr}p_{r1} & \cdots & \sum_r p_{mr}p_{rm} \end{pmatrix}$$

$$= \begin{pmatrix} p_{11}^{(2)} & \cdots & p_{1m}^{(2)} \\ \vdots & \ddots & \vdots \\ p_{m1}^{(2)} & \cdots & p_{mm}^{(2)} \end{pmatrix}$$

$$= P^{(2)}$$

$$\therefore P^{(2)} = P^2$$

Thus, 2-step transition probability matrix is the square of 1-step transition probability matrix.

n -step transition probabilities in matrix notation

It can be shown that

$$P^{(n)} = P^n$$

That is, n -step transition probability matrix is the n th power of the 1-step transition probability matrix.

Exercise:

Let $S = \{1, 2, 3\}$ and $P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}$.

- (a) Calculate the 2-step transition probability matrix.
- (b) Determine the values of (i) $P(X_2 = 3 | X_0 = 1)$ (ii) $P(X_2 = 2 | X_0 = 3)$
 (iii) $P(X_2 = 3 | X_0 = 2)$ (iv) $P(X_{12} = 3 | X_{10} = 1)$ (v) $P(X_{12} = 3 | X_{11} = 1)$.

Solution:

(a)

$$P^{(2)} = P^2 = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} = \begin{pmatrix} 0.12 & 0.31 & 0.57 \\ 0.14 & 0.32 & 0.54 \\ 0.13 & 0.32 & 0.55 \end{pmatrix}.$$

(b)

(i) $P(X_2 = 3 | X_0 = 1) = 0.57$

(ii) $P(X_2 = 2 | X_0 = 3) = 0.32$

(iii) $P(X_2 = 3 | X_0 = 2) = 0.54$

(iv) $P(X_{12} = 3 | X_{10} = 1) = 0.57$

(v) $P(X_{12} = 3 | X_{11} = 1) = 0.70$

Marginal distribution

Let the marginal distribution of X_n be denoted by $\underline{a}^{(n)} = (a_1^{(n)} \ a_2^{(n)} \ \cdots \ a_m^{(n)})$. Then, $\underline{a}^{(0)} = (a_1^{(0)} \ a_2^{(0)} \ \cdots \ a_m^{(0)})$ is the distribution of X_0 , that is, the initial distribution of the process. For simplicity, we can write $\underline{a}^{(0)} = \underline{a} = (a_1 \ a_2 \ \cdots \ a_m)$.

The j th element of the marginal distribution of X_n can be obtained as:

$$\begin{aligned}
a_j^{(n)} &= P(X_n = j) \\
&= \sum_i P(X_0 = i, X_n = j) \\
&= \sum_i P(X_0 = i) P(X_n = j \mid X_0 = i) \\
&= \sum_i a_i^{(0)} P_{ij}^{(n)}
\end{aligned}$$

That is, $\underline{a}^{(n)} = \underline{a}^{(0)} P^{(n)} = \underline{a} P^n$.

Example

There are three brands in the market: 1, 2 and 3. Customer preferences are observed at the beginning of each month. The initial distribution of customer preference is $\underline{a} = (0.2 \ 0.3 \ 0.5)$. The TPM is:

$$P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

- (a) Calculate the marginal distribution of X_1 .
- (b) What percent of customers used Brand 2 in Month 1?
- (c) What is the probability that a customer will use Brand 1 initially and Brand 3 in Month 1? (What percent of customers used Brand 1 initially and Brand 3 in the next month?)
- (d) Calculate $P(X_1 = 3, X_2 = 1, X_3 = 2)$.
- (e) Calculate the marginal distribution of X_{60} .
- (f) Calculate the marginal distribution of X_{61} .

Solution

$$(a) \underline{a}^{(1)} = \underline{a}^{(0)} P^{(1)}$$

$$\begin{aligned} &= (0.2 \ 0.3 \ 0.5) \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} \\ &= (0.13 \ 0.31 \ 0.56). \end{aligned}$$

$$(b) a_2^{(1)} = 0.31. \text{ Therefore, 31\% customers used Brand 2 in Month 1.}$$

$$(c) P(X_0 = 1, X_1 = 3)$$

$$= P(X_0 = 1) P(X_1 = 3 \mid X_0 = 1)$$

$$= a_1 P_{13}$$

$$= 0.2 \times 0.7$$

$$= 0.14$$

$$(d) P(X_1 = 3, X_2 = 1, X_3 = 2)$$

$$= \sum_i P(X_0 = i, X_1 = 3, X_2 = 1, X_3 = 2)$$

$$= \sum_i a_i P_{i3} P_{31} P_{12}$$

$$= 0.1 \times 0.2 \times \sum_i a_i P_{i3}$$

$$= 0.02 \times (0.2 \times 0.7 + 0.3 \times 0.4 + 0.5 \times 0.6)$$

$$= 0.0112$$

(There is an easier way to solve this problem. Try to find it.)

$$(e) \underline{a}^{(60)} = \underline{a}^{(0)} P^{(60)} = \underline{a} P^{60}.$$

(Do it by using R.)

$$(f) \underline{a}^{(61)} = \underline{a}^{(0)} P^{(61)} = \underline{a} P^{61}. \text{ (Think of an easier way.)}$$

Compare $\underline{a}^{(60)}$ and $\underline{a}^{(61)}$. Any comment?

Limiting behavior of a stochastic process

By using R code for the brand-switching example discussed earlier, we get

$$P^{(\infty)} = \begin{pmatrix} 0.132 & 0.319 & 0.549 \\ 0.132 & 0.319 & 0.549 \\ 0.132 & 0.319 & 0.549 \end{pmatrix} \text{ and } \underline{a}^{(\infty)} = (0.132 \ 0.319 \ 0.549).$$

This means that the market share reaches equilibrium (steady state) after a while. That is, market shares do not change with time, though individual customers still switch brands according to matrix P .

Also, all the rows of $P^{(\infty)}$ are identical and same as the steady-state marginal distribution (limiting distribution). This means that the steady-state market share does not depend on the initial distribution. It only depends on the one-step transition probabilities.

How to obtain the limiting distribution mathematically

Let us denote the limiting distribution by $\underline{\pi} = (\pi_1 \ \pi_2 \ \cdots \ \pi_m)$. That is, $\underline{\pi} = \underline{a}^{(\infty)}$. We can use the following formula to obtain $\underline{\pi}$ mathematically:

$$\underline{\pi} P = \underline{\pi}$$

where, $\pi_1 + \pi_2 + \cdots + \pi_m = 1$ and $\pi_i \geq 0$.

Example:

$$\text{Let } P = \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix}.$$

Obtain the limiting distribution $\underline{\pi}$.

Solution:

$$\underline{\pi} P = \underline{\pi}$$

$$\therefore (\pi_1 \ \pi_2 \ \pi_3) \begin{pmatrix} 0.1 & 0.2 & 0.7 \\ 0.2 & 0.4 & 0.4 \\ 0.1 & 0.3 & 0.6 \end{pmatrix} = (\pi_1 \ \pi_2 \ \pi_3)$$

which leads to the following equations:

$$0.1 \pi_1 + 0.2 \pi_2 + 0.1 \pi_3 = \pi_1$$

$$0.2 \pi_1 + 0.4 \pi_2 + 0.3 \pi_3 = \pi_2$$

$$0.7 \pi_1 + 0.4 \pi_2 + 0.6 \pi_3 = \pi_3$$

subject to

$$\pi_1 + \pi_2 + \pi_3 = 1 \text{ and } \pi_i \geq 0.$$

Solving the above equations, we get $\pi_1 = 0.132$, $\pi_2 = 0.319$ and $\pi_3 = 0.549$.

The two-armed bandit problem

The name is a reference to a type of gambling machine called two-armed bandit. The two arms offer different rewards, and the gambler has to decide which arm to play without knowing which arm is better.

A doctor is experimenting with two different treatments A and B, and does not yet know which treatment is better. A series of patients will each be given one of the two treatments. The doctor wants to find a strategy that ensures that as many as possible of the patients get the better treatment – even though we do not know which treatment is better!

Let us consider the following two strategies:

Random strategy: Allocate each patient to treatment A or B at random, each with probability 0.5.

Two-armed bandit strategy: For the first patient, choose A or B at random with equal probability. If patient n is given treatment A and it is successful, we use treatment A again for patient $n + 1$, $n = 1, 2, 3, \dots$. If A is a failure for patient n , we switch to treatment B for patient $n + 1$. The rule is similar if patient n is given treatment B.

Comparison of the two strategies

Our purpose is to determine which of the above two strategies is better. For treatment A, let the probability of success be denoted by α (unknown). For treatment B, the probability of success is β (unknown).

State space of the process:

$$\{AS, AF, BS, BF\}$$

where ‘AS’ means that the patient was given A and it was a success, and so on.

For random strategy:

$$P(A S) = P(A) P(S | A) = 0.5 \alpha$$

$$P(A F) = P(A) P(F | A) = 0.5 (1 - \alpha)$$

$$P(B S) = P(B) P(S | B) = 0.5 \beta$$

$$P(B F) = P(B) P(F | B) = 0.5 (1 - \beta)$$

Thus, for random strategy, the overall probability of success of a patient, denoted by p_R , is given by

$$p_R = P(A S) + P(B S) = 0.5 (\alpha + \beta)$$

For two-armed bandit strategy:

The TPM is:

$$\begin{array}{c} \begin{array}{cc} AS & AF & BS & BF \end{array} \\ \begin{array}{l} AS \\ AF \\ BS \\ BF \end{array} \begin{pmatrix} \alpha & 1 - \alpha & 0 & 0 \\ 0 & 0 & \beta & 1 - \beta \\ 0 & 0 & \beta & 1 - \beta \\ \alpha & 1 - \alpha & 0 & 0 \end{pmatrix} \end{array}$$

The limiting probabilities for each of the states can be obtained by using the formula discussed earlier. The overall probability of success of a patient, denoted by p_T , is obtained as

$$p_T = \frac{\alpha + \beta - 2\alpha\beta}{2 - \alpha - \beta}$$

Comparison: It can be shown that $p_T \geq p_R$.